

Python File Handling & Automation

Submitted By:
Manasa Sangadala

Objective:
To understand Python file handling, automation logic, and exception handling using Python programming.

Tools Used:

- Python
- Replit
- OS Module
- Shutil Module

Project Description:
This project demonstrates Python file handling and automation operations. The program performs different file operations such as creating files, writing data into files, reading file content, renaming files, and moving files into another folder.

The project also uses exception handling using try-except blocks to manage errors efficiently.

Python Code:

```
import os
import shutil

try:
    file = open("students.txt", "w")
    file.write("Manasa\nRavi\nKiran")
    file.close()

    print("Data written successfully")

    file = open("students.txt", "r")
    content = file.read()
    print("\nFile Content:")
```

```
print(content)
file.close()
```

```
os.rename("students.txt", "student_data.txt")
print("\nFile renamed successfully")
```

```
if not os.path.exists("backup_folder"):
    os.mkdir("backup_folder")
```

```
shutil.move("student_data.txt", "backup_folder/student_data.txt")
print("File moved successfully")
```

```
except Exception as e:
    print("Error:", e)
```

Output:

Data written successfully

File Content:

Manasa

Ravi

Kiran

File renamed successfully

File moved successfully

Features Implemented:

- File creation
- File writing
- File reading
- File renaming
- File moving
- Exception handling
- Automation operations

Conclusion:

Successfully implemented Python file handling and automation operations using Python.

```
Python Creator (1) | Upgrade | Preview | main.py | Shell | + | Search | Invite | Publish | Format

main.py | ...
1 import os
2 import shutil
3
4 try:
5     file = open("students.txt", "w")
6     file.write("Manasa\nRavi\nKiran")
7     file.close()
8
9     print("Data written successfully")
10
11     file = open("students.txt", "r")
12     content = file.read()
13     print("\nFile Content:")
14     print(content)
15     file.close()
16
17     os.rename("students.txt", "student_data.txt")
18     print("\nFile renamed successfully")
19
20     if not os.path.exists("backup_folder"):
21         os.mkdir("backup_folder")
22
23     shutil.move("student_data.txt", "backup_folder/student_data.txt")
24     print("File moved successfully")
25
26 except Exception as e:
27     print("Error:", e)
28
```

```
Python Creator (1) | Upgrade | Preview | main.py | Shell | + | Search | Invite | Publish | Format

~/workspace/python main.py
- ~/workspace/python main.py
+ Data written successfully

File Content:
Manasa
Ravi
Kiran

File renamed successfully
File moved successfully
~/workspace
```